

ANKUR LAHIRY

CONTACT INFORMATION

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RESEARCH INTEREST

High Performance Computing, Machine Learning, Graph Representation Learning, Decision Making, Anomaly Detection, Synthetic Performance Data Generation

EDUCATION

Texas State University

Ph.D. in Computer Science, Department of Computer Science

CGPA: 3.92 / 4.00; Expected Graduation: Summer 2027

Fall 2022 – Present

San Marcos, Texas

Bangladesh University of Engineering and Technology (BUET)

Undergraduate, Department of Computer Science and Engineering, CGPA: 3.00 / 4.00

May 2012 – February 2017

Dhaka, Bangladesh

PUBLICATIONS

- [1] Ankur Lahiry, Banooqa Banday, Yugesh Bhattarai, Tanzima Z Islam, and Mohammad Zaeed. “COMPASS: A Unified Decision-Intelligence System for Navigating Performance Trade-off in HPC”. In: *arXiv preprint arXiv:2604.22688* (2026).
- [2] Ankur Lahiry, Banooqa Banday, Yugesh Bhattarai, and Tanzima Z Islam. “WANDER: An Explainable Decision-Support Framework for HPC”. In: *arXiv preprint arXiv:2506.04049* (2025).
- [3] Ankur Lahiry, Ayush Pokharel, Banooqa Banday, Seth Ockerman, Amal Gueroudji, Mohammad Zaeed, Tanzima Z Islam, and Line Pouchard. “A Distributed Framework for Causal Modeling of Performance Variability in GPU Traces”. In: *arXiv preprint arXiv:2510.18300* (2025).
- [4] Ankur Lahiry, Ayush Pokharel, Seth Ockerman, Amal Gueroudji, Line Pouchard, and Tanzima Z Islam. “Scalable GPU Performance Variability Analysis framework”. In: *arXiv preprint arXiv:2506.20674* (2025).
- [5] Chase Phelps, Ankur Lahiry, Tanzima Z Islam, and Line C Pouchard. “Reimagine Application Performance as a Graph: Novel Graph-Based Method for Performance Anomaly Classification in High-Performance Computing”. In: *2024 IEEE 48th Annual Computers, Software, and Applications Conference (COMPSAC)*. IEEE. 2024, pp. 240–245.
- [6] Chase Phelps, Ankur Lahiry, Tanzima Z Islam, and Christopher Kelly. “Graph Based Anomaly Detection in Chimbuko: Feasible or Fallible?”. In: *Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis (SC '23)* (2023).
- [7] Tarek Ramadan, Ankur Lahiry, and Tanzima Z Islam. “Novel representation learning technique using graphs for performance analytics”. In: *2023 International Conference on Machine Learning and Applications (ICMLA)*. IEEE. 2023, pp. 1311–1318.

RESEARCH EXPERIENCE

• Doctoral Research Assistant, Texas State University (Spring 2023 – Present)

- Developed a unified decision-intelligence system for HPC environments to recommend performant configurations by balancing speed, cost, and reliability trade-offs with explainable outputs and uncertainty-aware ranking, scaling to traces with 1.3B samples (126 GB) and achieving up to 100× faster training and 80× faster inference than state-of-the-art generative baselines
- Design graph-based and explainable AI methods that turn complex system logs into intuitive signals, enabling early detection and diagnosis of unusual performance behavior in large-scale HPC systems.
- Engineered scalable GPU log-analysis pipelines for large trace datasets (multiple large-scale GPU traces per experiment), using distributed partitioning and parallel processing to reduce analysis time and memory overhead, and achieving a 67% improvement in scalability while enabling fast identification of performance variability, memory stalls, and system bottlenecks across repeated HPC runs
- Collaborate with BNL and ANL to align methods with real production supercomputing workflows, improving the reliability and efficiency of scientific computing at scale.

- **Undergraduate Research, Bangladesh University of Engineering and Technology (2016 – 2017)**
 - Conducted undergraduate research on improving the efficiency and stability of the **Simplex Algorithm** for linear programming optimization problems.
 - Proposed and implemented a novel degeneracy-resolution approach that outperformed the conventional **Bland's Rule** in computational efficiency while preserving solution accuracy.

INTERNSHIP EXPERIENCE

- **Summer Intern, Brookhaven National Laboratory, Upton, New York (June 2023 – August 2023)**

Worked as a research intern on the **Chimbuko Project**, a performance analytics framework used to monitor and improve the efficiency of large-scale supercomputing applications.

 - Developed a **novel representation learning approach** to automatically detect performance anomalies in complex computing workflows.
 - Analyzed real performance data from high-performance computing systems to identify inefficiencies and unusual behaviors.
 - Helped improve system reliability by designing techniques that enable faster detection and interpretation of abnormal performance events.

TECHNICAL SKILLS

- **Model Architectures:** Graph Neural Networks (GNN), Deep Neural Networks (DNN), Large Language Models (LLMs), Generative AI (Gen AI), Recommendation Systems.
- **Framework:** PyTorch, HuggingFace
- **Core Techniques:** Representation Learning, Anomaly Detection, Explainable AI (XAI), Feature Engineering, Transfer Learning, Retrieval-Augmented Generation (RAG).
- **Applications:** Performance Analytics in HPC, Decision Making, System Behavior Modeling, Anomaly Classification, Intelligent Monitoring Systems, Synthetic Data Generation.

TEACHING EXPERIENCE

- **Doctoral Instructional Assistant, Texas State University** *Fall 2022*

Assisted in teaching **Computer Architecture, CS4310** courses, supporting lectures and lab sessions.

 - Guided undergraduate students through assignments, lab work, and term projects.
 - Collaborated with faculty to refine lab materials and evaluate course assignments.
 - Encouraged active learning and technical discussions to improve student engagement.

PROFESSIONAL EXPERIENCE

- **Software Engineer, DeshiPay (DataBird)** *Dhaka, Bangladesh | 2022*

Led development of secure mobile payment systems and scalable iOS infrastructure.

 - Developed real-time financial transaction features for mobile payment platforms.
 - Collaborated with cross-functional teams on scalable and secure system design.
- **Software Engineer, Ridmik Labs (DataBird)** *Dhaka, Bangladesh | 2020 – 2022*

Contributed to the development of large-scale real-time communication platforms.

 - Developed messaging and VoIP systems using WebRTC and XMPP.
 - Improved application scalability, reliability, and user experience.
- **Senior Software Engineer, Prefeex Ltd.** *Dhaka, Bangladesh | 2019 – 2020*

Led development of scalable reservation platform architecture and mobile applications.

 - Designed and implemented core application modules and system architecture.
 - Improved scalability and development efficiency through Agile practices.
- **Software Engineer, iPay Systems Ltd.** *Dhaka, Bangladesh | 2017 – 2019*

Developed secure digital payment applications and reusable SDK frameworks.

 - Implemented security and validation mechanisms for financial transactions.
 - Built reusable payment modules to support third-party integrations.
- **Senior Software Engineer, BellBizzer Inc.** *Seattle, WA, USA (Remote)*

Developed scalable iOS architecture for a rental marketplace platform.

 - Collaborated with product and design teams on system scalability and usability.
 - Optimized application performance and long-term maintainability.